

Straightening Laws in Algebraic Complexity

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Section 1

Background

What is Algebraic Complexity?

Question

How hard is it to compute polynomials?

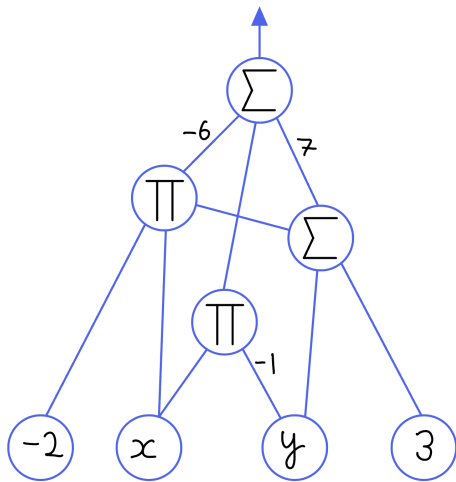
Question

How hard is it to compute polynomials in restricted models?

Question

Can we find *explicit* polynomials which are hard to compute?

Algebraic Circuits



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

Horner's Rule

Hard Polynomials Exist

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Instead of working over a field $\mathbb{F}$, we take a parameter ε , allow our coefficients to be rational functions in ε , and let ε tend to 0.

Definition

We say a circuit *border computes* $g(\bar{x})$ over $\mathbb{F}$ if it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}(\varepsilon)[\bar{x}], \quad q \geq 1.$$

We **cannot** set $\varepsilon = 0$ since we may divide by ε .

What does Border Computation Get Us?

Question

What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

Border Waring Rank

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the *circuit complexity* of some family of polynomials.

Definition

Fix some polynomials $g_1(\bar{x}), \dots, g_k(\bar{x}) \in \mathbb{F}[\bar{x}]$.

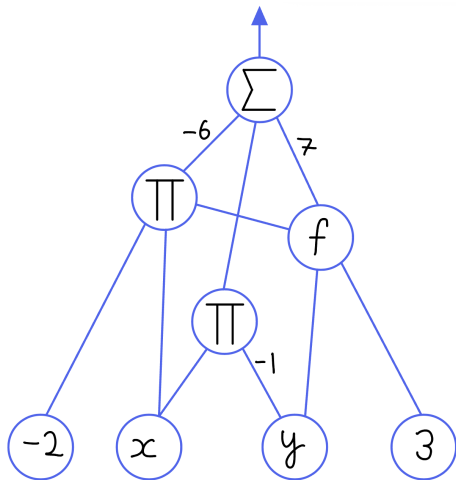
The *ideal* generated by $g_1(\bar{x}), \dots, g_k(\bar{x})$ is the set of combinations

$$\langle g_1, \dots, g_k \rangle = \left\{ \sum_{i=1}^k h_i(\bar{x}) \cdot g_i(\bar{x}) \mid h_i(\bar{x}) \in \mathbb{F}[\bar{x}] \right\}.$$

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

Algebraic Circuits with Oracles



Question

How can we compare the *relative* complexity of two polynomials?

We can allow a fixed polynomial $f(\bar{x})$ to be computed with unit cost by allowing it to function as its own gate.

Figure: $-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$

Principal Ideals

Example

The *principal ideals* are generated by a single polynomial g .

If $f \in \langle g \rangle$, then g is a *factor* of f .

Question

Suppose $f \in \langle g \rangle$. Does g have a small f -oracle circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If g is a factor of f , $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *border computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

Question

If $f(\bar{x})$ has a small circuit, do $g_1(\bar{x})$ or $g_2(\bar{x})$ have a small f -oracle circuit?

This is open for *general* polynomials $g_1(\bar{x}), g_2(\bar{x})$ [Gro20].

Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator. We can instead ask about ideals whose generators have *additional structure*.

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Prior Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = n^{\Theta(1)}$.

What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = n^{\Theta(1)}$.

- There exist polynomial size circuits for computing $\det(X)$.
- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Interpolation* as well as a new application of the *Isolation Lemma*.

Section 2

Preliminaries

Bideterminants

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$. Consider the *bitableau* $(S \mid T)$ of shape σ

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right).$$

Then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{pmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{pmatrix} \begin{pmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{pmatrix} (x_{4,3}).$$

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^d x_{r_i, c_i}$ is given by a bideterminant:

$$\prod_{i=1}^d x_{r_i, c_i} = \left(\begin{array}{c|c} \boxed{r_1} & \boxed{c_1} \\ \boxed{r_2} & \boxed{c_2} \\ \boxed{\vdots} & \boxed{\vdots} \\ \boxed{r_d} & \boxed{c_d} \end{array} \right) (X).$$

Thus, the bideterminants span $\mathbb{F}[X]$.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B} (A \mid B)(X),$$

where A and B are *conjugate semistandard*.

Since $I_{n,m,r}^{\det}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r .

Proposition ([BC03, Corollary 1.7])

The ideal $I_{n,m,r}^{\det}$ has as basis the conjugate semistandard bideterminants of shape σ whose first row has length at least r .

Shuffle Products

$$\begin{aligned}
 & \left(\begin{array}{|c|c|c|c|c|c|} \hline \overset{\bullet}{i_1} & \dots & \overset{\bullet}{i_p} & i_{p+1} & \dots & i_s \\ \hline \end{array} \mid \begin{array}{|c|c|c|} \hline j_1 & \dots & j_s \\ \hline \end{array} \right) \left(\begin{array}{|c|c|c|c|c|c|} \hline \overset{\bullet}{\ell_1} & \dots & \overset{\bullet}{\ell_q} & \ell_{q+1} & \dots & \ell_b \\ \hline \end{array} \mid \begin{array}{|c|c|c|} \hline m_1 & \dots & m_b \\ \hline \end{array} \right) \\
 &= \sum_{\sigma} \text{sgn}(\sigma) \left(\begin{array}{|c|c|c|c|c|c|} \hline \sigma(i_1) & \dots & \sigma(i_p) & i_{p+1} & \dots & i_s \\ \hline \end{array} \mid \begin{array}{|c|c|c|} \hline j_1 & \dots & j_s \\ \hline \end{array} \right) \left(\begin{array}{|c|c|c|c|c|c|} \hline \sigma(\ell_1) & \dots & \sigma(\ell_q) & \ell_{q+1} & \dots & \ell_b \\ \hline \end{array} \mid \begin{array}{|c|c|c|} \hline m_1 & \dots & m_b \\ \hline \end{array} \right)
 \end{aligned}$$

Where the summation is over all permutations σ such that $\sigma(i_1) < \dots < \sigma(i_p)$ and $\sigma(\ell_1) < \dots < \sigma(\ell_q)$.

Laplace Expansion

$$\begin{aligned}
 & \left(\begin{array}{c|c} \overset{\bullet}{i_1} & \overset{\bullet}{i_p} \end{array} \begin{array}{c} \vdots \\ \vdots \end{array} \begin{array}{c} i_{p+1} \\ \vdots \\ i_s \end{array} \mid \begin{array}{c} j_1 \\ \vdots \\ j_s \end{array} \right) \left(\begin{array}{c|c} \overset{\bullet}{\ell_1} & \overset{\bullet}{\ell_q} \end{array} \begin{array}{c} \vdots \\ \vdots \end{array} \begin{array}{c} \ell_{q+1} \\ \vdots \\ \ell_t \end{array} \mid \begin{array}{c} m_1 \\ \vdots \\ m_t \end{array} \right) \\
 &= \sum_{\tau, \mu} \text{sgn}(\tau) \text{sgn}(\mu) \left(\begin{array}{c|c} i_1 & \ell_1 \end{array} \begin{array}{c} \vdots \\ \vdots \end{array} \begin{array}{c} i_p \\ \vdots \\ i_q \end{array} \mid \begin{array}{c} \tau(j_1) \\ \vdots \\ \tau(j_p) \end{array} \begin{array}{c} \mu(m_1) \\ \vdots \\ \mu(m_q) \end{array} \right) \\
 & \quad \left(\begin{array}{c|c} i_{p+1} & \tau(i_{p+1}) \end{array} \begin{array}{c} \vdots \\ \vdots \end{array} \begin{array}{c} i_s \\ \vdots \\ \tau(i_s) \end{array} \mid \begin{array}{c} \ell_{q+1} \\ \vdots \\ \ell_s \end{array} \mid \begin{array}{c} \mu(m_{q+1}) \\ \vdots \\ \mu(m_t) \end{array} \right)
 \end{aligned}$$

Where the summation is over all permutations σ such that $\sigma(i_1) < \dots < \sigma(i_p)$ and $\sigma(\ell_1) < \dots < \sigma(\ell_q)$.

Example Straightening

Consider the following nonstandard tableau

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right)$$

This is not conjugate semistandard in the first two rows, so we focus on straightening these first two rows.

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|} \hline \dot{2} & \dot{3} \\ \hline \dot{1} & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) + \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right)$$

Laplace:

$$\begin{aligned} \left(\begin{array}{|c|c|} \hline \dot{2} & \dot{3} \\ \hline \dot{1} & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) &= \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 1 & 2 \\ \hline 3 & & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline 1 & & \\ \hline \end{array} \right) \\ &= - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right) \end{aligned}$$

We notice that our unstraightened rows appear in the first expression so that we can solve for it.

Example Straightening

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right)$$

Thus, we get that

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline 2 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline 1 & & \\ \hline \end{array} \right)$$

The first term is conjugate semistandard, but not the next two terms.

All terms that appear have the same content and they are either the same shape with lexicographically smaller filling or they have a longer shape.

Main Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = n^{\Theta(1)}$.

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Main Theorem

Theorem

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(n^2 m^2 d^3 (n^3 + m^3))$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

Let $\sigma = (4, 2, 1)$:

$$(K_\sigma \mid K_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \right) (X)$$

= product of upper-left minors.

Section 3

Proof Techniques

Proof Outline

Since we are starting with $f(X) \in I_{n,m,r}^{\det} \subseteq \mathbb{F}[X]$, we can express in terms of our standard basis:

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X).$$

Our goal will be to use $f(X)$ as an oracle to compute a canonical bideterminant.

To do this, we will isolate one of the terms $(S_k \mid T_k)(X)$.

Linear Operators

Definition (Substitution operator)

For $i < j$, the operator $\text{Sub}_{i \rightarrow j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j , replacing that i with j , and sorting the row in increasing order.

$$\text{Sub}_{2 \rightarrow 4} \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 & & \\ \hline 1 & & & \\ \hline \end{array}$$

Let $h_i^j(S)$ be the number of times i is replaced by j in S .

Linear Operators

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

In fact, after enough applications we arrive at *anticanonical* tableau:

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n . Then

$$\left(\begin{array}{c} \text{Sub} \\ n-1 \rightarrow n \end{array} \circ \begin{array}{c} \text{Sub} \\ n-2 \rightarrow n \end{array} \circ \begin{array}{c} \text{Sub} \\ n-2 \rightarrow n-1 \end{array} \circ \cdots \right. \\ \left. \circ \begin{array}{c} \text{Sub} \\ 2 \rightarrow n \end{array} \circ \cdots \circ \begin{array}{c} \text{Sub} \\ 2 \rightarrow 3 \end{array} \circ \begin{array}{c} \text{Sub} \\ 1 \rightarrow n \end{array} \circ \cdots \circ \begin{array}{c} \text{Sub} \\ 1 \rightarrow 2 \end{array} \right)(S) = \overline{K}_\sigma.$$

Applying the Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}$, adding a multiple of row j to row i .

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau. Then we have that

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

For $0 \leq h \leq h_i^j(S) - 1$, let $\mathcal{C}_{i \rightarrow j}^h(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing.

Applying all Substitution Operators

We have seen now that after applying a sequence of substitution operators $\text{Sub}_{i \rightarrow j}$, we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Translating this to the setting of linear operators $E_{i,j}(\lambda_{i,j})$ gives the following:

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (\overline{K}_{\sigma_k} \mid \overline{K}_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Applying all Substitution Operators

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (\bar{K}_{\sigma_k} \mid \bar{K}_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Let J_n be the $n \times n$ matrix with 1's along the anti-diagonal.

Lemma

$$(\bar{K}_\sigma \mid \bar{K}_\sigma)(J_n X J_m) = \pm (K_\sigma \mid K_\sigma)(X).$$

Applying all Substitution Operators

Lemma ([DG25, Corollary 3.10])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MJ_nXJ_mN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Next, we introduce another variable v , and modify D and D' as follows:

$$\begin{aligned}\tilde{D} &= \text{diag}(y_1 v, y_2 v^2, \dots, y_n v^n) \in \mathbb{F}[Y, v]^{n \times n} \\ \tilde{D}' &= \text{diag}(z_1 v, z_2 v^2, \dots, z_m v^m) \in \mathbb{F}[Z, v]^{m \times m}\end{aligned}$$

Lemma

Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(\tilde{D} X \tilde{D}') = Y^{\bar{s}} Z^{\bar{t}} v^{|S|+|T|} \cdot (S \mid T)(X),$$

where $(\bar{s}, \bar{t})$ is the multidegree of $(S \mid T)(X)$.

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_n \tilde{D} X \tilde{D}' J_m N) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X),$$

Linear transformations just correspond to a layer of summation gates giving us an f -oracle circuit for the above.

Use *interpolation* to get an f -oracle circuit computing $\text{coeff}_{v_{d_{\min}}}(g)$.

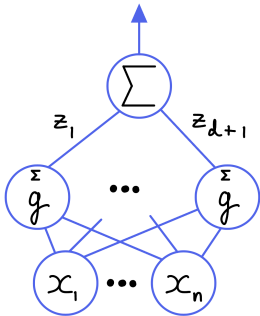
Interpolation

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.



Then, we can compute $\text{coeff}_{t_i}(g)$ by a size $O(ds)$, depth Δ f -oracle circuit.

Proof of Interpolation: Interpolation

Let $\alpha_1, \dots, \alpha_{d+1}$ be $d + 1$ distinct elements in $\mathbb{F}$.

$$\begin{pmatrix} \alpha_1^d & \alpha_1^{d-1} & \cdots & \alpha_1 & 1 \\ \alpha_2^d & \alpha_2^{d-1} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_d^d & \alpha_d^{d-1} & \cdots & \alpha_d & 1 \\ \alpha_{d+1}^d & \alpha_{d+1}^{d-1} & \cdots & \alpha_{d+1} & 1 \end{pmatrix} \begin{pmatrix} \text{coeff}_{t^d}(f) \\ \text{coeff}_{t^{d-1}}(f) \\ \vdots \\ \text{coeff}_t(f) \\ \text{coeff}_1(f) \end{pmatrix} = \begin{pmatrix} f(\bar{x}, \alpha_1) \\ f(\bar{x}, \alpha_2) \\ \vdots \\ f(\bar{x}, \alpha_d) \\ f(\bar{x}, \alpha_{d+1}) \end{pmatrix}$$

This left matrix is a *Vandermonde* matrix and is invertible since the α_i are all distinct.

Proof of Interpolation: Interpolation

$$\begin{pmatrix} \text{coeff}_{t^d}(f) \\ \text{coeff}_{t^{d-1}}(f) \\ \vdots \\ \text{coeff}_t(f) \\ \text{coeff}_1(f) \end{pmatrix} = \begin{pmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,d} & z_{1,d+1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,d} & z_{2,d+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{d,1} & z_{d,2} & \cdots & z_{d,d} & z_{d,d+1} \\ z_{d+1,1} & z_{d+1,2} & \cdots & z_{d+1,d} & z_{d+1,d+1} \end{pmatrix} \begin{pmatrix} f(\bar{x}, \alpha_1) \\ f(\bar{x}, \alpha_2) \\ \vdots \\ f(\bar{x}, \alpha_d) \\ f(\bar{x}, \alpha_{d+1}) \end{pmatrix}$$

$$\text{coeff}_{t^i}(f) = \sum_{j=1}^{d+1} z_{d+1-i,j} f(\bar{x}, \alpha_j).$$

Where Does Isolation Arise?

We now have an f -oracle circuit to compute

$$\sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

We want an f -oracle circuit for a single $i \in I$.

Our proof shows that for each $k \in I$, the tuple $(\bar{e}_k, \bar{f}_k, \bar{s}_k, \bar{t}_k)$ is distinct.

Interpolation is *insufficient* for our purposes.

Issues with Interpolation

If a circuit border computes $g(\bar{x})$, then it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon), \quad q \geq 1.$$

Idea: Interpolate with respect to ε .

Problem: q can be *exponentially large*

$\implies$ Interpolation gives *large circuit*.

Idea: Kronecker Substitution

Say there is a specific coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$ we want to *isolate*.

Lemma

Suppose that $\deg(g) \leq d$, then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

Problem: The resulting polynomial has *large degree*.

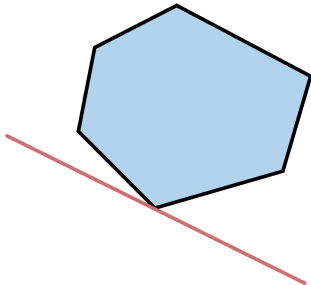
$\implies$ Interpolation gives *large circuit*.

Isolation Lemma

Say there is a specific coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e}} c_{\bar{e}} \bar{x}^{\bar{e}}$ we want to *isolate*.

Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.



Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

Isolation Lemma

Lemma ([KS01, Lemma 4])

Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

Let $z_1, \dots, z_\ell$ be independently and uniformly chosen from $\{0, \dots, M\}$ at random, where $M \geq K\ell/\varepsilon$.

Then, with probability at least $1 - \varepsilon$, there is a unique form in $\mathcal{L}$ of minimum value at $z_1, \dots, z_\ell$.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be choose to be small
 $\implies$ unique lowest $\deg_w$ -term and interpolation w.r.t w is small.

Section 4

Related and Future Work

Pfaffian Analogues

Definition

If X is a $2n \times 2n$ skew-symmetric matrix, then $\det(X)$ is the perfect square of another polynomial called the *Pfaffian* of X .

Theorem ([DG25, Theorem 1.6])

Let $f \in I_{2n, 2r}^{\text{pfaff}}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, \deg(f))$ that *exactly computes* the $t \times t$ pfaffian for some $t = \Theta(r^{1/3})$.

Symmetric Analogue?

Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} := x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix.

Conjecture

Let $I_{n,r}^{\text{sym}} \subseteq \mathbb{F}[X]$ be the ideal generated by the $r \times r$ minors of X . Let $f \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial. Then there exists a constant-depth f -oracle circuit of size $\text{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a symmetric matrix for $t = \text{poly}(n)$.

Symmetric Basis

Definition (d-tableau, d-determinant)

Following [Con94], we say a bitableau $(S \mid T)$ of shape σ and its associated bideterminant $(S \mid T)(X)$ are *d-tableau* and *d-bideterminants* respectively if S and T are both conjugate semistandard Young tableau and if furthermore the single tableau formed by stacking alternating rows of S and T ,

$$\begin{array}{|c|} \hline S_1 \\ \hline T_1 \\ \hline \vdots \\ \hline \vdots \\ \hline S_k \\ \hline T_k \\ \hline \end{array}$$

is conjugate semistandard.

Straightening for Symmetric Determinantal Ideals

Theorem ([dCP76, Theorem 5.1], [Con94, Theorem 2.2])

Let $(S \mid T)$ be a bitableau of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A \mid B)} c_{A,B} (A \mid B)(X),$$

where the $(A \mid B)(X)$ are d-bideterminants of shape $\tau \geq \sigma$, and $c_{A,B} \in \mathbb{Z}$.

Partial Progress

Theorem

Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial, d the total degree of f , and $f(X) = \sum_{k \in [s]} \alpha_k (S_k \mid T_k)(X)$ the expansion of $f(X)$ in the d -bideterminant basis where $(S_k \mid T_k)$ is of shape σ_k and each $\alpha_k \in \mathbb{F}$ is nonzero. Then there exists a partition σ and a depth-three f -oracle circuit of size $O(d^3 n^7)$ computing $\alpha[K_\sigma](X)$ for some nonzero $\alpha \in \mathbb{F}$.

Our main roadblock for fully resolving the symmetric case is the reduction from a product of determinants to a single determinant.

Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

Question

Is there an analogue to our results for the ideal of permanents of a matrix?

The main roadblock is that we have no analogue of the following result:

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

Algebras with Straightening Laws

Definition ([Eis80])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra. A is an *Algebra with a Straightening Law* if

- A is a free R -module with a basis given by the standard monomials
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^m r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

These relations in the second requirement are known as the *straightening relations* or *straightening laws*.

Hankel Ideals

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \leq i \leq n$. For $1 \leq j \leq n$, let X_j be the following Hankel matrix:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

For $1 \leq r \leq \min \{ j, n - j + 1 \}$ let $I_r(X_j)$ be the ideal generated by $r \times r$ minors of X_j .

Hankel Ideals

The ideals $I_r(X_j)$ do not have the structure of an algebra with a straightening law.

In particular, the product of standard monomials need not be standard.

We still conjecture the following:

Conjecture

Let $I_r(X_j) \subseteq \mathbb{F}[x_1, \dots, x_n]$ be the ideal generated by the $r \times r$ minors of X_j . Let $f \in I_r(X_j)$ be a nonzero polynomial. Then there exists a constant-depth f -oracle circuit of size $\text{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a Hankel matrix for $t = \text{poly}(n)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

This no longer works:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix} = \begin{pmatrix} x_1 + \lambda x_2 & x_2 + \lambda x_3 & x_3 + \lambda x_4 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix}$$

Thank You!

Algebra is generous; she often gives more than is asked of her.

— Jean Le Rond d'Alembert

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